

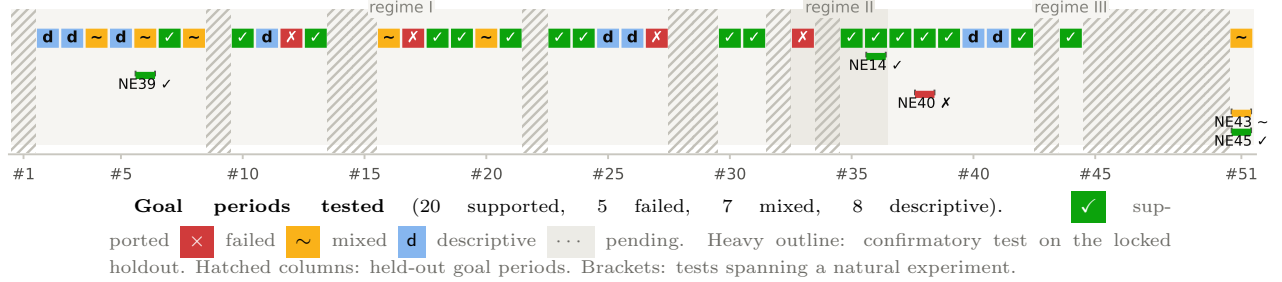
H74: An operator-grade change detector

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 3/5

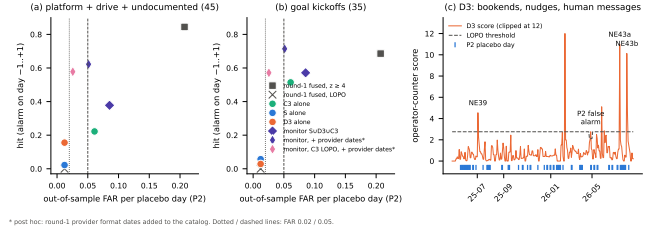
Bottom line. Calibrated out of sample, schema diff plus log-count operator counters plus topic shift is a usable monitor for drive, format and goal changes (FAR 0.05–0.085); scaffold changes stay invisible.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H74 builds a daily change detector for an agent swarm and scores it against 217 dated changes on 282 non-reserved days (April 2025 to September 2026). Round 1: a five-channel fused alarm was too noisy (18% false alarms on quiet days); a schema diff of the log records was precise. **Round 2** keeps three channels: the schema diff S , operator counters $D3$ (bookends, nudges, human messages) and $H36$'s frozen topic rule $C3$. Thresholds are calibrated on quiet days of the *other* goal periods, so every false-alarm rate (FAR) is out-of-sample. The monitor has FAR 0.085 [0.04, 0.17] per quiet day and hits goal kickoffs at 0.57 (p 0.003) and room changes at 6/6. $D3$ dates all three undocumented drive steps (NE39, NE43a, NE43b) at FAR 1/82. Scaffold tool changes are hit at chance. Verdict: **mixed**.



Observables. (a, b) Operating points: out-of-sample FAR per quiet day against hit rate (alarm on day $-1..+1$), for 45 platform, drive and undocumented events and for 35 goal kickoffs. Round 1's fused alarm (gray square) sits at FAR 0.21; with a calibrated threshold it fires once (gray cross). The monitor (violet) reaches FAR 0.085; pink and the small violet marker add round 1's provider-format dates to the catalog (post hoc). (c) The $D3$ counter channel over all days with its threshold (2.1–2.8): it fires at NE39 and both NE43 steps, and on one quiet day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Models 02 and 11. A day d is one step of a driven system. A platform step changes record schemas for every agent at once; an operator step changes drive counters; a goal step is a field step on content. Counts are heavy-tailed, so each counter is logged, scored by a robust trailing z (10 days) and must persist for two days. Each channel k alarms above $\theta_k(g)$, the conformal $(1 - \alpha)$ quantile of its scores on the n placebo days outside goal period g (placebo: ≥ 2 active days from every catalogued event).

$$p_f(d) = \min\{|z_f(d-1)|, |z_f(d)|\}, \quad z_f = z[\log(1 + x_f)], \quad D3(d) = \max_f p_f(d)$$

$$\theta_k(g) = s_{k, ([1-\alpha](n+1))} \text{ over placebo days } \notin g, \quad \alpha = 0.02$$

$$\text{alarm}(d) = [z_S > \theta_S] \vee [D3 > \theta_D] \vee [R1 \geq 3 \vee Z_{\text{cont}} \geq 2]$$

H74-change-detector · generated 2026-10-04 21:04 by in-
fra/summaries/build_summaries.py · card hypotheses/H74-change-
detector/README.md · ratings: H74 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
r1 P1 tool changes AUC ≥ 0.70 , $p < 0.05$	AUC 0.68, p 0.059	mixed
r1 P4 fused FAR ≤ 0.10	0.18	failed
r2 P1a channel FAR ≤ 0.03 ; union ≤ 0.06	S 0.012, D3 0.012, C3 0.061; 0.085	partly
r2 P1b class hits in predicted bands	goal 0.57, drive 0.57, tool 0.35, held room 1.00, roster 0.26	
r2 P1c $p < 0.05$ goal; drive; tool $p < 0.10$	0.003; 0.13; 0.55	partly
r2 P1e beats C3 alone by ≥ 0.15	pooled hit 0.22 $\rightarrow$ 0.38 at held +0.024 FAR	
r2 R1 kill rule (operator-grade)	FAR 0.085; tool p 0.55	mixed
r2 P3.1 NE39 alarms (heavy-tailed baseline)	quantile score no; amended D3 yes (4.5 > 2.76)	held (amended)
r2 P3.3 NE43a/b alarm	amended D3: both	held
r2 P2.1 NE40 by presence rule (recovery)	primary 2 days late; one-day and appearance rules on the day; 0–1/37 FAR	mixed
r2 P4 provider DiD	0/25 after Holm; minimum p 0.031	unpowered

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 ~ mixed
G05 d descriptive
G06 ~ mixed
G07 ✓ supported
G08 ~ mixed
G10 ✓ supported
G11 d descriptive
G12 × failed
+30 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The pre-registered placebo pool held two provider format changes that round 1 found but did not catalog; they blind S (threshold 20). The catalog completion that fixes it is post hoc (FAR 0.051). The D3 channel and the accumulating presence rule are amendments after the synthetic study, before real data; synthetic power for count steps is 0.4–0.6. NE40 is a recovery, not a discovery. With dense events, random dates near other changes set a high chance level (0.33). The reserved data is not run.

ROUND 2

Re-freeze `confirm.py` on the round-2 monitor and run it on the reserved data after sign-off. Split the schema diff into platform and provider streams. Find an observable for scaffold changes (quantile-baselined mix: post hoc lead, p 0.016). Power the provider DiD with agent-day permutations.

- **Essence.** a cheap monitor that tells an operator, on the day, that the platform, the drive or the goal changed, with a known false-alarm rate.
- **H74-R1** Precision monitor.
- **H74-R2** Presence/retirement rule for formats.
- **H74-R3** Heavy-tailed baselines.
- **H74-R4** Provider API changes as natural experiments.
- **H74-R5** Holdout.

Second figure. Round 1. Left: AUC of the event window against placebo windows, by class and channel. Each channel sees its own class: schema and drive for platform and operator steps, topic for goals and rooms. Right: #51 daily channel scores; the schema channel fires on the search-tool rename (NE45), the drive channel on the bookend and nudger stops (NE43).